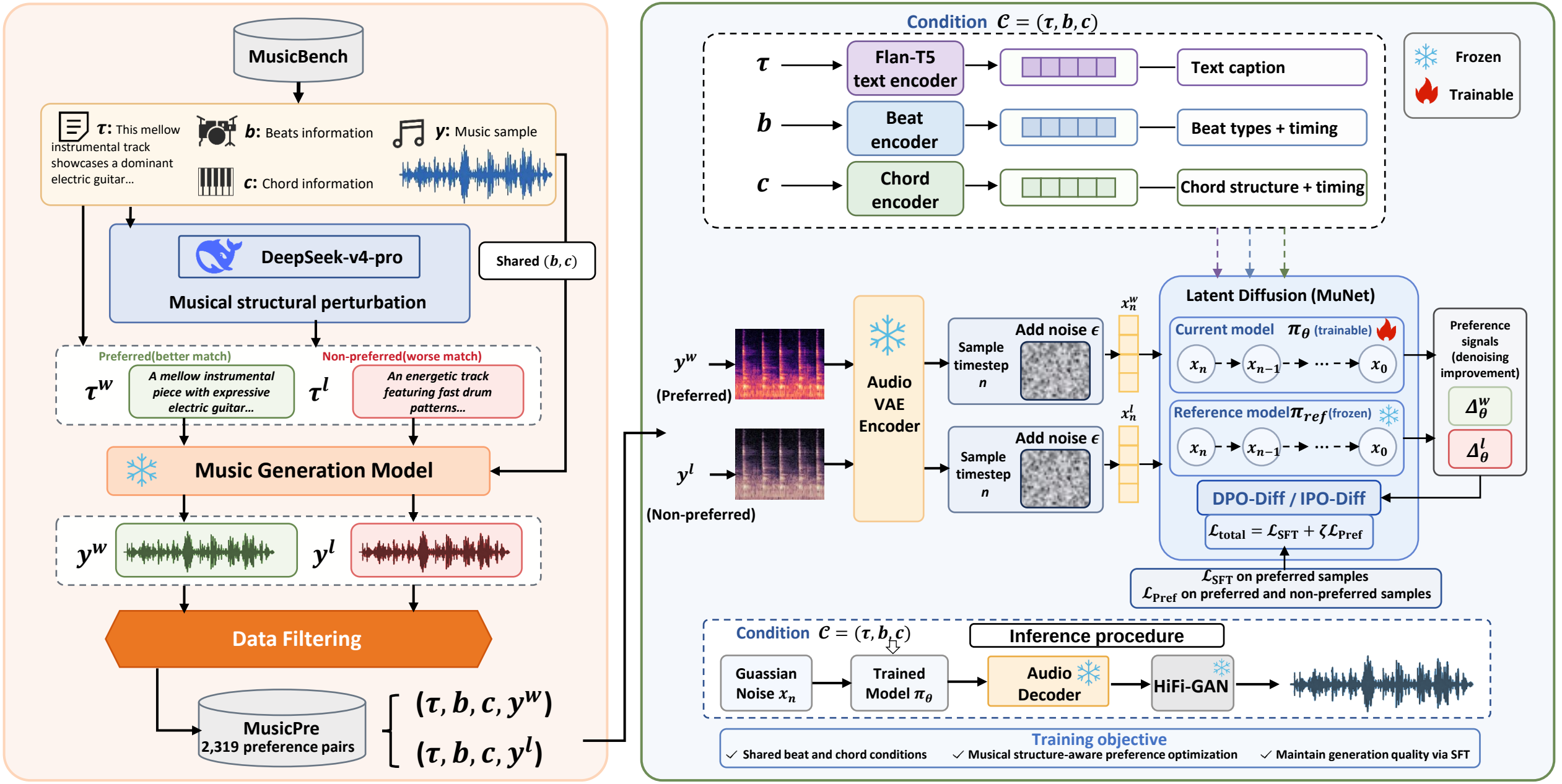

MusicBench
τ: This mellow instrumental track showcases a dominant electric guitar...
b: Beats information
c: Chord information
y: Music sample
DeepSeek-v4-pro
Musical structural perturbation
Shared (b, c)
Preferred(better match)
Non-preferred(worse match)
τ^w
A mellow instrumental piece with expressive electric guitar...
τ^l
An energetic track featuring fast drum patterns...
Music Generation Model
y^w
y^l
Data Filtering
MusicPre
2,319 preference pairs
(τ, b, c, y^w)
(τ, b, c, y^l)
Structure-Guided Preference Construction
Condition C = (τ, b, c)
τ
Flan-T5 text encoder
Text caption
b
Beat encoder
Beat types + timing
c
Chord encoder
Chord structure + timing
Frozen
Trainable
y^w
(Preferred)
y^l
(Non-preferred)
Audio VAE Encoder
Sample timestep n
Add noise ε
x_n^w
x_n^l
Latent Diffusion (MuNet)
Current model π_θ (trainable)
Reference model π_ref (frozen)
x_n
x_{n-1}
x_0
Preference signals (denoising improvement)
Δ_θ^w
Δ_θ^l
DPO-Diff / IPO-Diff
L_total = L_SFT + ζL_Pref
L_SFT on preferred samples
L_Pref on preferred and non-preferred samples
Condition C = (τ, b, c)
Inference procedure
Guassian Noise x_n
Trained Model π_θ
Audio Decoder
HiFi-GAN
Training objective
✓ Shared beat and chord conditions
✓ Musical structure-aware preference optimization
✓ Maintain generation quality via SFT
Preference optimization training process